

Summary

Large-scale wildlife conservation requires effective decisions under limited personnel, monitoring capacity, and geographic accessibility. In this study, we develop a spatial decision-making framework for Etosha National Park that integrates conservation value, threat exposure, accessibility, and resource availability to determine where protection efforts should be prioritized.

First, we divide the park into management units and identify the main conservation challenges, including poaching and illegal entry, human-wildlife conflict, wildfire, and water-related environmental stress. Wildlife populations, important habitats, water sources, historical threat information, and accessibility conditions are converted into normalized indicators. These indicators are combined to estimate the relative protection demand of each management unit. The highest-priority regions are identified as **[Region(s)]**, with priority scores ranging from **[value]** to **[value]**.

We then construct a resource allocation model for rangers and drone-assisted monitoring. Rather than treating the two resources as equally effective, we distinguish between monitoring and response functions. Drones provide efficient large-area surveillance but require personnel to operate them and have limited direct intervention capability, while ground personnel remain essential for verification, deterrence, and physical response. The model therefore allocates both personnel time and drone operating time while accounting for travel and accessibility constraints.

Protection effectiveness is evaluated using a combined monitoring-response service score. Under the proposed allocation strategy, the overall protection score reaches **[score]**, compared with **[baseline score]** under the reference allocation. The required staffing level to maintain the selected protection standard is estimated at **[number]** personnel under the baseline scenario.

Finally, sensitivity and scenario analyses are conducted by varying personnel availability, drone capacity, threat weights, and operational assumptions. The results show that **[main robust finding]**, while **[main sensitive factor]** has the greatest influence on the final allocation.

Overall, the proposed framework provides a practical and adaptable method for prioritizing conservation needs and allocating limited resources in large protected areas where complete surveillance is infeasible.

Keywords: Wildlife Conservation; Spatial Resource Allocation; Multi-Criteria Assessment; Reachability Analysis; Optimization; Etosha National Park

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1. Introduction

Wildlife conservation in large protected areas is fundamentally a problem of decision-making under limited resources. National parks often contain diverse habitats, uneven wildlife distributions, and multiple natural and human-induced threats, while the number of rangers, monitoring devices, and patrol opportunities remains limited. Effective conservation therefore requires not only identifying threats, but also determining where protection is most urgently needed and how limited resources should be deployed.

Etosha National Park in Namibia provides a representative case. Covering approximately 22,935 km², the park contains savannas, grasslands, salt pans, an extensive road network, and numerous natural and artificial water sources. It supports ecologically important wildlife, including endangered black rhinoceros, while facing pressures such as poaching, unauthorized entry, wildfire, and water stress. Because conservation value, threat exposure, and accessibility vary substantially across the park, uniform protection is neither practical nor efficient.

To address this spatial heterogeneity, we divide Etosha into management units and evaluate each unit using two main dimensions: conservation value and threat exposure. Conservation value reflects the importance of wildlife populations, habitats, and key ecological features, while threat exposure represents the relative pressure from human and natural hazards. These factors are combined to estimate protection demand. Geographic accessibility and travel conditions are then incorporated separately so that high-priority areas can be compared not only by need, but also by the practicality of reaching and protecting them.

2. Assumptions and Notations

Symbol	Definition
x_i	Evaluation indicator
$\tilde{x}_i$	Standardized indicators
μ_i	Average value
s_i	Standard deviation
R	Correlation coefficient matrix
y_i	Principal Components
b_i	The information contribution of eigenvalue
T	Composite score

3. Model

3.1. Major Challenges and Priority Levels

The ordering above is provisional and is based on threat evidence, potential losses, spatial extent, and intervention feasibility. Because event and loss data are not available under a common measurement standard, this ranking cannot be interpreted as an empirically measured ordering of risk. Regional

Tentative Priority	Challenge	Basis for Assessment
1	Poaching and illegal entry	The problem statement explicitly identifies continuing rhino-poaching pressure. Illegal losses of priority species are difficult to replace and can be mitigated through patrol and law-enforcement interventions.
2	Harmful wildfires and habitat destruction	The problem statement indicates that wildfires can affect large areas of the park and may simultaneously threaten animals, plants, and habitats. Prescribed burns must be distinguished from harmful fires.
3	Rainfall anomalies and water-source pressure	Water sources influence animal aggregation and habitat use, while rainfall anomalies indirectly affect conservation needs through water availability and fuel conditions.

Table 1: Tentative management priorities.

conditions, seasonal conditions, and events that have already occurred may change the ordering. Human–wildlife conflict, disease, and other threats remain pending items and are not classified as low-risk merely because data are unavailable.

3.2. Considered Factors and Quantitative Treatment

The data are organized into 17 regions: 15 animal-survey statistical regions, the main salt-pan region, and other areas outside the survey. Animal, ecological, and road data originate from 2015, 2021, and 2026, respectively, and are used as the planning baseline.

Animal responsibility and habitat responsibility are quantified separately:

$$V_i^A = \frac{\sum_s w_s (n_{is}/N_s)}{\sum_s w_s}, \quad N_s = \sum_{i \in I_S} n_{is}; \quad V_i^F = \frac{A_i^F}{\sum_i A_i^F}. \quad (1)$$

Here, n_{is} denotes the abundance of species s in region i , I_S denotes the surveyed area, and A_i^F denotes the potentially burnable habitat area. The category weights for huntable/protected/specially protected species are provisionally set to 1/2/3 as a model assumption. Abundance and density are not double-weighted. Black-rhino conflict counts are temporarily excluded from the main table but the species remains a priority conservation target. Survey-external and missing values remain unknown. A zero fuel baseline for the salt pan does not imply zero ecological value. If only a 12-month multi-year rainfall average is available, an actual interannual anomaly cannot be calculated.

3.3. Evaluation of Whether Protection Is Effective

A target is considered “protected” when priority targets can be effectively patrolled and monitored, anomalies can be verified, qualified teams can respond in a timely manner, and necessary water-source

Factor	Data Processing and Quantification	Model Meaning
Area and spatial partition	Use a unified coordinate system and repair overlaps and gaps. Animal density uses the original survey area, while spatial operations use geometric area.	Basis for defining the assessment scope and evaluating input per unit area.
Animal abundance and protection category	Associate abundance with species and survey regions; first calculate each species' regional share, then apply protection-category weights.	Regional animal conservation responsibility V_i^A .
Vegetation and important habitats	Divide land-cover area by the effective classified area to obtain proportions. The grassland/shrubland/forest area share represents potentially burnable-habitat responsibility, while important plant locations are recorded separately.	Habitat structure and fire-side conservation responsibility V_i^F .
Roads and access conditions	Convert the shortest entry time into $E = \exp(-T^E/\tau_E)$; travel time from protection stations to targets is calculated separately.	Human-entry accessibility and response reachability; not a probability of poaching.
Historical burned area	After quality-assurance screening, deduplicate late-dry-season burned areas year by year; divide burned area by effective burnable area and then take the multi-year mean.	Fire exposure F_i , not the probability of ignition or harmful loss.
Rainfall	Compare precipitation accumulated over the preceding complete rainy season with a fixed climatic baseline: $R = (P - \bar{P})/\bar{P}$.	Background condition for water-source pressure and fire season; no assumption is made that less rainfall necessarily causes more fire.
Water sources	Verify coordinates, type, and seasonal operating status. For reliable water points, take the union of buffers and divide by regional area.	Reliable water-source coverage W_{it} and locations for priority monitoring.

Table 2: Factors considered and their quantitative treatment.

and ecological monitoring is maintained.

For the same target, threat, and time period, let $d \in [0, 1]$ represent effective patrol/monitoring coverage and verification capability. In the baseline model, response capability r is binary: $r = 1$ if a qualified team can arrive within the selected response limit, and $r = 0$ otherwise. The protection service score is

$$g = dr. \quad (2)$$

The score is first calculated at the target point and then aggregated according to protection demand. How resource inputs are converted into detection and response capability is determined in the subsequent resource-allocation model.

$$D_{ikt} = \theta_k V_i^k H_{ikt}, \quad (3)$$

$$P_t = 100 \frac{\sum_{i,k} D_{ikt} g_{ikt}}{\sum_{i,k} D_{ikt}}, \quad (4)$$

$$L_t = \sum_{i,k} D_{ikt} (1 - g_{ikt}). \quad (5)$$

Here, D represents relative protection demand, H represents the corresponding threat exposure, and θ represents the management weight. P is the protection-capability score, while L is unmet protection demand. On the anti-poaching side, E is used as an exposure proxy; on the fire side, F is used as an exposure proxy. Rainfall and water sources are treated separately as environmental conditions and auxiliary evaluation factors.

Evaluation Level	Criterion
Model-internal capability	Under the same demand conditions, P increases and L decreases, while minimum service levels for priority targets and required ecological-monitoring requirements are satisfied. Specific thresholds are determined in the resource model.
Actual animal protection	Illegal-animal loss rates and priority-population status are evaluated using comparable surveys and event records. An increase in detected events does not automatically indicate worse protection.
Actual habitat and water-source protection	Evaluate the impact of harmful unplanned fires on important habitats, the status of reliable water sources, and key ecological conditions. Total burned area must not automatically be treated as protection failure.

Table 3: Protection effectiveness at different evaluation levels.

The model score measures protection capability; actual effectiveness must be validated separately using ecological and event data while controlling for monitoring effort, seasonality, and environmental differences. When key data are unknown, a complete score is not reported. A park-wide average score cannot substitute for minimum requirements imposed on priority targets.

3.4. Data Sources and Computation

The source materials and calculations are based on the following files and analyses:

- [Original problem statement](#);
- [Species-source verification](#);
- [Ecological data processing](#);
- [Fire-data processing plan](#);
- [Rainfall-factor design](#);
- [Question 1 baseline calculations](#).